David Awoyemi

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RESEARCH PROJECTS

Evidence-Based Learning *Innovations*

Three signature research projects — presented as instructional design case studies with problem context, theoretical frameworks, methodology, evaluation, outcomes, and critical reflection.

01

IMMERSIVE LEARNING TECHNOLOGY RESEARCH

Development of Evidence-Based Immersive Virtual Reality-Supported Construction Safety Engineering Training

Dissertation

Virtual Reality · AI-Enhanced Learning · Engineering Education · Behavioral Analytics · Microgenetic Analysis · AERA 2026

PROBLEM CONTEXT

STEM learners — particularly civil engineering students — consistently struggle to transfer abstract safety and hazard concepts into applied, authentic professional contexts. Traditional classroom instruction and static simulations fail to replicate the cognitive demands of real-world hazard identification, resulting in knowledge gaps that have serious professional

LEARNERS & CONTEXT

- Undergraduate civil engineering students
- Limited access to authentic hazard training environments
- High conceptual and cognitive load in safety training
- Diverse prior knowledge and technical proficiency levels
- Need for transferable, applied decision-making skills

DESIGN FRAMEWORKS

ADDIE — structured the systematic design, development, and evaluation phases.

Cognitive Load Theory (Sweller, 1988) — informed the scaffolding of intrinsic, extraneous, and germane load within the VR environment.

Experiential Learning Theory (Kolb, 1984) — grounded the design in concrete experience, reflective observation, and active experimentation.

Design and Development Research (DDR) — enabled iterative refinement of the intervention based on participant data.

METHODS & TOOLS

VR Simulation Platform | Tobii Pro Eye-Tracking | EmoBit Physiological Sensors | Microgenetic Analysis | Sequential Behavior Mining | Bayesian Cognitive Diagnostics | Mixed Methods

DESIGN SOLUTION

responded to learner behavior in real time, adjusting the level of scaffolding based on performance patterns identified through behavioral mining algorithms. Metacognitive prompts were embedded at decision points to encourage self-regulation and reflection.

EVALUATION & OUTCOMES

📊 **Pre/post assessments showed measurable improvements in hazard identification accuracy and response confidence**

🧠 **Behavioral analytics revealed distinct learning state patterns associated with effective versus ineffective regulation strategies**

🏆 **Featured in Vanguard Nigeria; research presented at AERA 2026 (2 accepted presentations), AECT 2025, and iLRN 2023**

DISSEMINATION

Awoyemi, I. D., et al. (AERA 2026, Accepted). Unpacking in-VR learning dynamics for heat stress training: A microgenetic and sequential behavior analysis. Roundtable.

Awoyemi, I. D., et al. (AERA 2026, Accepted). Feedback-driven strategic shifts in immersive VR training: A multi-phase behavioral-mining study. Poster.

Moon, J., Awoyemi I. D., & Abu S. (Accepted). Decoding learning states in VR safety training: A Bayesian cognitive diagnostic approach. *Behaviourmetrika*.

CRITICAL REFLECTION

"This project transformed my understanding of what evaluation can mean in instructional design. Moving beyond self-report surveys to embed physiological and behavioral data collection into the learning environment revealed a richness of evidence I had not previously considered. Future iterations will further personalize AI-generated feedback

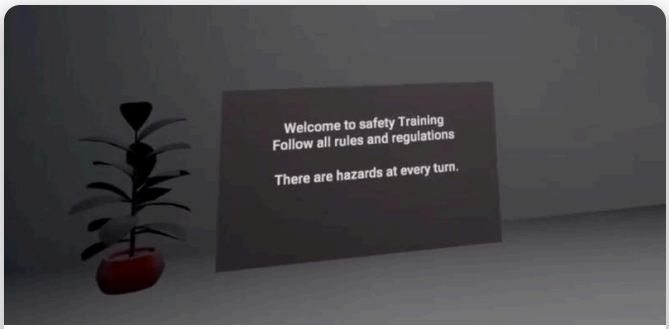
VISUAL EVIDENCE

IVR Prototype — Phase 1 & Phase 2 Visuals

PHASE 1 OF 2

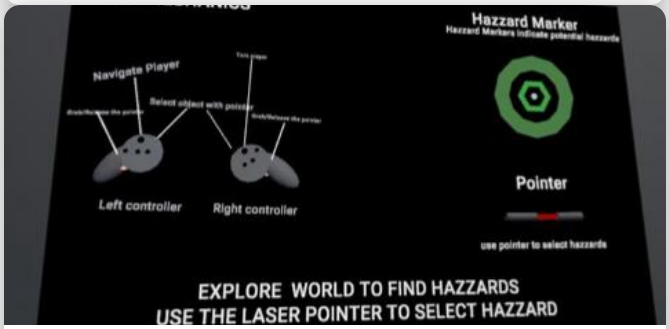
Hazard Identification Scenarios

Five immersive construction and industrial environments — learners navigate to hazard marker locations to identify risks across **Caught-In, Falling Objects, Slip & Fall, Unstable Scaffolding,** and **Electrical Hazards.** Grounded in Experiential Learning Theory (Kolb, 1984) and Situated Learning Theory (Lave & Wenger, 1991).



Reception / Onboarding

Falling Objects — Scenario 2



Game Mechanics

Caught-In Between — Scenario 1

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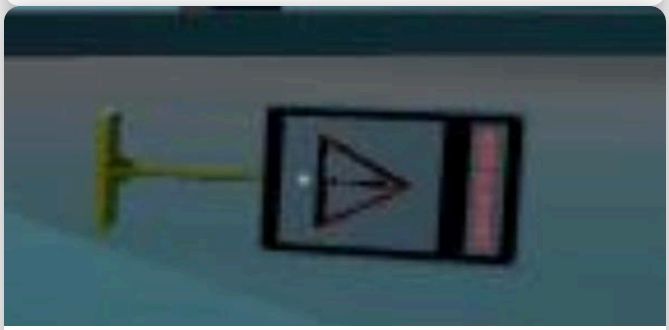
🏗️ Construction Site Overview

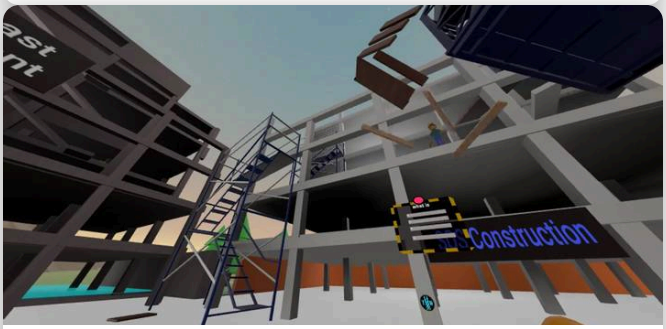
🏗️ Site Environment



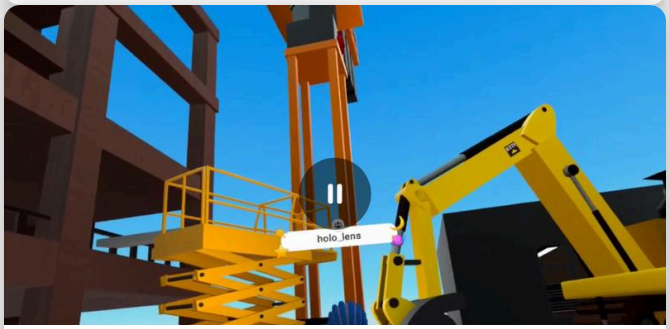
💧 Slip & Fall — Scenario 3

👷 SOS Construction — Multi-Avatar

📱 Mobile Feedback Screen



🏗️ SOS Construction — Wide View



⚙️ Caught-In Hazard — Equipment Zone

🏗️ Caught-In Scenario — Full Site

🏛️ Reception Hall — Full View

⚡ Gulf Coast — Slippery Surface

🌙 Night Scaffold — Hazard Markers

💧 Gulf Coast — Wet Surface Zone

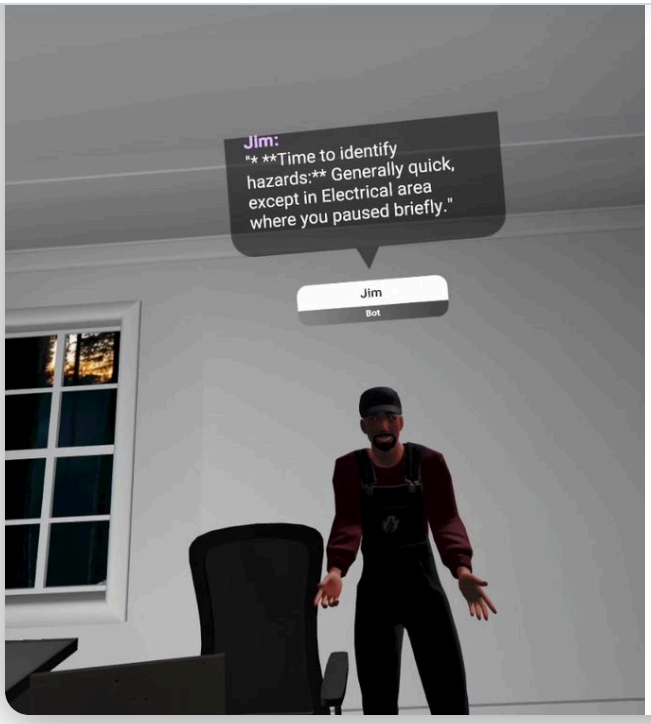
🌙 Night Scaffold — Close-Up

🚊

PHASE 2 OF 2

AI Avatar Feedback & Adaptive Coaching

Jim — the AI worker avatar — delivers personalized two-stage feedback: first **diagnosing** specific performance gaps, then guiding learners through **corrective procedures** step by step, mirroring the role of an experienced field supervisor.

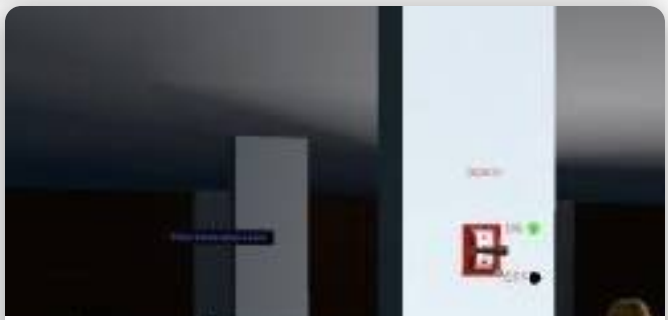


Jim — VR Diagnostic & Feedback Agent

Jim is an AI-powered worker avatar deployed in the virtual debrief room. After each scenario, Jim delivers **diagnostic feedback** identifying performance gaps, followed by **corrective guidance** walking learners through proper hazard-resolution procedures — modeling the role of an experienced field supervisor.

Spoken Dialogue | Diagnostic Feedback | Corrective Guidance | Mastery-Gated

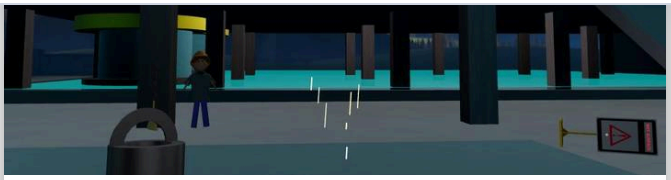
AI Avatar — Diagnostic Phase: Worker avatar with hard hat positio



Lockout/Tagout — Power Panel Close-Up: Interactive ON/OFF switch

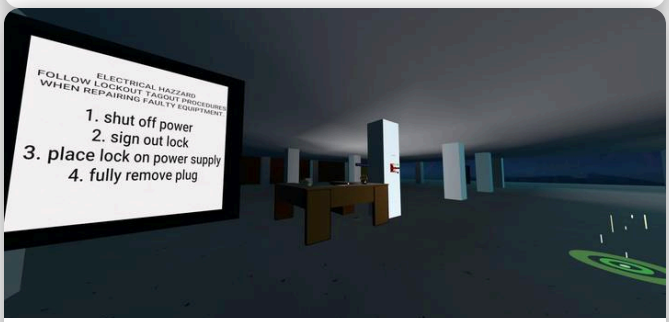
LOTO Training Step 1 — Power Isolation: AI avatar guides learner

LOTO Corrective Feedback Board — 4-Step Procedure: AI avatar pres



Corrective Feedback — Electrical Hazard Room (Wide View): AI avat

Jim (AI Avatar) — Diagnostic Check-In: Avatar asks the learner wh



Corrective Feedback — Gulf Coast Slip Hazard: AI avatar presents

Jim (AI Avatar) — Skills Assessment:

Slip-Hazard Remediation — Wet Floor Sign Placement: AI avatar pos

Jim (AI Avatar) — Performance Debrief:

02

AI LITERACY CURRICULUM DESIGN

AI-WISE: Industry-Informed Artificial Intelligence Literacy Curriculum for Higher Education

University of Alabama · College of Education Rising Tide Grant · 2024–2025 · Laura McNeill (PI)

AI Literacy · Curriculum Design · Higher Education · Industry Partnership · Design-Based Research · AECT 2025

Undergraduate students routinely graduate without the practical, ethical, and critical AI literacy skills that industry employers expect. Existing curricula treat AI as a technical topic rather than a multi-dimensional professional and ethical competency. There is a significant gap between what universities teach about AI and what the workforce requires — a gap that disproportionately affects students from underrepresented groups who may have less informal AI exposure.

STAKEHOLDERS & CONTEXT

- Undergraduate students across disciplines
- Industry partners providing competency frameworks
- Faculty collaborators across departments
- College of Education as institutional home
- AY 2024–2025 implementation timeline

DESIGN FRAMEWORKS

Design-Based Research (DBR) — iterative co-design with industry and faculty partners.

Backward Design (Wiggins & McTighe) — outcomes-first curriculum architecture aligned to industry competency frameworks.

Universal Design for Learning (UDL) — modular, multimodal curriculum structure accessible across disciplines and learning contexts.

Culturally Responsive Pedagogy — explicit attention to representation, equity, and the social dimensions of AI.

METHODS & TOOLS

Industry Competency Mapping | Focus Groups | Pilot Testing | Survey Instruments

DESIGN SOLUTION

I co-developed AI-WISE (AI Workforce Integration and Skill Enhancement) — a modular undergraduate curriculum that integrates **practical AI skills, ethical reasoning, and reflective learning** informed by direct industry consultation. The curriculum was designed to be discipline-agnostic, enabling faculty across departments to embed AI literacy into existing courses without requiring technical expertise. Modules addressed: AI fundamentals, prompt engineering, ethical implications, bias and fairness, AI in professional contexts, and critical evaluation of AI-generated content.

EVALUATION & OUTCOMES

✅ **Piloted with undergraduate students; feedback highlighted relevance, clarity, and cross-disciplinary applicability**

📣 **Presented at AECT 2025 (Las Vegas), UCF Teaching & Learning with AI Conference (May 2025), and ITLC-Lilly (Accepted, 2026)**

CRITICAL REFLECTION

"Designing AI literacy curriculum for a diverse undergraduate population required me to think beyond technical content and engage with the social, ethical, and identity dimensions of AI. One lesson I will carry forward: faculty development is not peripheral to curriculum success — it is its precondition. Scaling AI-WISE will require institution-wide professional development, infrastructure support, and ongoing co-design with the students whose learning lives are most directly shaped by AI technologies."

03

LEARNING ANALYTICS & PHYSIOLOGICAL COMPUTING

Beyond Self-Report: Using Eye-Tracking and Physiological Data to Evaluate

Physiological Computing Eye-Tracking Learning Analytics

Elementary Education Broadening Participation AERA 2025

PROBLEM CONTEXT

Traditional evaluation in instructional design relies heavily on self-report surveys, post-hoc assessments, and behavioral observation — methods that capture only a thin slice of the learning experience. Cognitive load, attentional focus, emotional engagement, and moment-to-moment processing remain largely invisible. This limitation is particularly acute when studying young learners (elementary students) who may lack the metacognitive vocabulary to accurately self-report their cognitive states.

LEARNERS & CONTEXT

- Upper elementary students (Grades 4–6)
- Skyland Elementary School, Tuscaloosa City Schools
- Diverse racial and socioeconomic backgrounds
- Code-N-Sensor Summer Camp context
- Physiological computing curriculum activities

DESIGN FRAMEWORKS

Mixed Methods Evaluation Design — integrating physiological, behavioral, and self-report data streams.

Cognitive Load Theory — guiding the interpretation of pupil dilation and fixation patterns as indicators of intrinsic and extraneous load.

Culturally Responsive STEM Education — informing curriculum design to affirm student identities and promote computing career aspirations.

METHODS & TOOLS

Tobii Pro Lab (Eye-Tracking) | EmoBit (Physiological Sensors) | Pupil Diameter Analysis | Fixation Pattern Analysis | Cross-Case Analysis | Thematic Coding | SPSS / R

DESIGN SOLUTION & CONTRIBUTION

I served as a Graduate Research Assistant and Camp Staff Team Member on this NSF ITEST project, contributing to curriculum design, data collection, analysis, and dissemination. My specific contributions included: co-designing physiological computing activities (heart rate monitoring, data collection exercises) that were both instructionally meaningful and suitable for data collection; developing coding schemes for behavioral and attentional analysis; and conducting cross-case analyses to identify patterns across diverse learner profiles. The project demonstrated the **feasibility and instructional value of integrating physiological data into formative evaluation** — providing evidence that pupil dilation reliably tracks cognitive load during problem-solving tasks in elementary students.

EVALUATION & OUTCOMES

🔬 **Published in ACM SIGCSE 2024 proceedings; additional manuscripts under review and accepted**

👧 **Novel insights into elementary girls' experiences in physiological computing — contributing to broadening participation in CS**

💰 **NSF ITEST grant of $1,125,666 awarded — validating the project's significance and scalability**

CRITICAL REFLECTION

children. It also deepened my appreciation for evaluation as an iterative, design-embedded practice rather than a summative add-on. Going forward, I intend to integrate physiological evaluation methods into my instructional design practice — not as research novelty, but as a more honest and responsive approach to understanding how learning is actually experienced."

View All Publications → ID Portfolio →

Idowu David Awoyemi

Ph.D. Candidate · Instructional Technology
University of Alabama · ADIE Lab

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David Awoyemi

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